

A DUALITY THEOREM

In this chapter, after establishing a number of preliminary results on Grothendieck topologies, we state and prove a duality theorem between the subtoposes of the classifying topos of a geometric theory and the *quotients* of the theory, i.e. the geometric theory extensions over its signature. We then discuss the proof-theoretic significance of the theorem, obtaining an alternative proof of it based on proof-theoretic ideas. We conclude the chapter with a deduction theorem for geometric logic analogous to the well-known one for classical first-order logic, which we establish by using the above-mentioned proof-theoretic interpretation of the duality theorem.

3.1 Preliminary results

3.1.1 A 2-dimensional Yoneda lemma

In this section we present a 2-dimensional version of the Yoneda lemma which will be useful in the sequel.

Recall that there are a number of 2-categories which naturally play a role in topos theory; among them, there are the 2-category $\mathbf{Cat}$ of small categories, functors and natural transformations between them and the 2-category $\mathbf{BTop}$ of Grothendieck toposes, geometric morphisms and geometric transformations between them. Also, we have all the 2-categories arising from notable fragments of geometric logic, namely the 2-category $\mathbf{Cart}$ of cartesian categories, cartesian functors and natural transformations between them, the 2-category $\mathbf{Reg}$ of regular categories, regular functors and natural transformations between them, the 2-category $\mathbf{Coh}$ of coherent categories, coherent functors and natural transformations between them, and the 2-category $\mathbf{Geom}$ of geometric categories, geometric functors and natural transformations between them.

Given a strict 2-category $\mathcal{R}$ and two 0-cells a and b in $\mathcal{R}$, we say that a and b are *equivalent* if there exist 1-cells $f : a \rightarrow b$ and $g : b \rightarrow a$ and invertible 2-cells $\alpha : f \circ g \Rightarrow 1_b$ and $\beta : g \circ f \Rightarrow 1_a$. Given a 2-category $\mathcal{R}$, we have an obvious 2-functor $Y : \mathcal{R} \rightarrow [\mathcal{R}^{\text{op}}, \mathbf{Cat}]$ (where $[\mathcal{R}^{\text{op}}, \mathbf{Cat}]$ is the 2-category of 2-functors $\mathcal{R}^{\text{op}} \rightarrow \mathbf{Cat}$), which sends a 0-cell a to the (obviously defined) 2-functor $Y(a) := \mathcal{R}(-, a) : \mathcal{R}^{\text{op}} \rightarrow \mathbf{Cat}$. Notice that the above notion of equivalence specializes in $\mathbf{Cat}$ to the well-known notion of equivalence between small categories.

The following result is well-known, but we present a proof for the reader's convenience.

Proposition 3.1.1 *With the above notation, for any $a, b \in \mathcal{R}$, the functors $Y(a)$ and $Y(b)$ are equivalent (as 0-cells in the 2-category $[\mathcal{R}^{\text{op}}, \mathfrak{Cat}]$) if and only if a and b are equivalent (as 0-cells in $\mathcal{R}$).*

Proof It is easy to see that two 2-functors $F, G : \mathcal{R}^{\text{op}} \rightarrow \mathfrak{Cat}$ are equivalent if and only if, for each $c \in \mathcal{R}$, the categories $F(c)$ and $G(c)$ are equivalent via functors $K(c) : F(c) \rightarrow G(c)$ and $L(c) : G(c) \rightarrow F(c)$, naturally in $c \in \mathcal{R}$, i.e. for any 1-cell $f : c \rightarrow d$ in $\mathcal{R}$ the obvious naturality squares for both K and L commute up to an invertible natural transformation.

Now suppose that for $a, b \in \mathcal{R}$ we have that $Y(a)$ and $Y(b)$ are equivalent via transformations $K : Y(a) \Rightarrow Y(b)$ and $L : Y(b) \Rightarrow Y(a)$ such that $K \circ L \cong Y(b)$ and $L \circ K \cong Y(a)$. Then we have $K(a) : \mathcal{R}(a, a) \rightarrow \mathcal{R}(b, a)$ and $L(b) : \mathcal{R}(b, b) \rightarrow \mathcal{R}(a, b)$; let us put $f := K(a)(1_a) : a \rightarrow b$ and $g := L(b)(1_b) : b \rightarrow a$. We want to prove that $g \circ f \simeq 1_a$ and $f \circ g \simeq 1_b$. Consider the naturality square for K corresponding to the arrow $g : b \rightarrow a$:

$$\begin{array}{ccc} \mathcal{R}(a, a) & \xrightarrow{K(a)} & \mathcal{R}(a, b) \\ \downarrow - \circ g & & \downarrow - \circ g \\ \mathcal{R}(b, a) & \xrightarrow{K(b)} & \mathcal{R}(b, b). \end{array}$$

By our hypothesis, this square commutes up to an invertible natural transformation, so $f \circ g = K(a)(1_a) \circ g \cong K(b)(g) \cong K(b)(L(b)(1_b)) \cong 1_b$. Dually, or more explicitly by replacing K with L and f with g in the above argument, one obtains an isomorphism $g \circ f \cong 1_a$. So the 1-cells f and g give an equivalence between a and b , as required.

The ‘if’ direction of the proposition is trivial. $\square$

3.1.2 An alternative view of Grothendieck topologies

In a category $\mathcal{C}$ we call a collection of arrows in $\mathcal{C}$ with common codomain a *presieve*; given a presieve P on $c \in \mathcal{C}$, we define the sieve $\overline{P}$ generated by P as the collection of arrows in $\mathcal{C}$ with codomain c which factor through an arrow in P .

Given a collection $\mathcal{U}$ of presieves on $\mathcal{C}$, we define the Grothendieck topology *generated by $\mathcal{U}$* as the smallest Grothendieck topology J on $\mathcal{C}$ such that all the sieves generated by the presieves in $\mathcal{U}$ are J -covering.

Given two Grothendieck topologies J and J' on a category $\mathcal{C}$ such that $J' \supseteq J$, we say that J' is *generated over J by a collection $\mathcal{U}$ of sieves in $\mathcal{C}$* if J' is generated by the collection of sieves on $\mathcal{C}$ which are either J -covering or belonging to $\mathcal{U}$.

Given a site $(\mathcal{C}, J)$ and a sieve S on an object c of $\mathcal{C}$, we denote by $\overline{S}^J$ the J -closure $\{f : \text{dom}(f) \rightarrow c \mid f^*(S) \in J(\text{dom}(f))\}$ of S ; notice that $\overline{S}^J$ is the smallest J -closed sieve on c containing S . Recall that, if $\mathcal{C}$ is small, via the identification of sieves on an object c with subobjects in $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ of $\mathcal{C}(-, c)$, $\overline{S}^J$

corresponds to the closure of $S \mapsto \mathcal{C}(-, c)$ with respect to the universal closure operation on $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ associated with the (local operator corresponding to the) Grothendieck topology J on $\mathcal{C}$ as in Theorem 1.3.36.

Remark 3.1.2 Given a functor $F : \mathcal{C} \rightarrow \mathcal{E}$, where $\mathcal{E}$ is a Grothendieck topos, and a presieve P in $\mathcal{C}$, F sends P to an epimorphic family if and only if it sends $\overline{P}$ to an epimorphic family.

We note that the classical definition of a Grothendieck topology (cf. Definition 1.1.1(c)) can also be put in the following alternative form (cf. the notion of pretopology in [3], vol. 269 II.1.3):

Definition 3.1.3 A Grothendieck topology on a category $\mathcal{C}$ is an assignment J sending any object c of $\mathcal{C}$ to a collection $J(c)$ of sieves on c in such a way that

- (a) the maximal sieve M_c belongs to $J(c)$;
- (b) for each pair of sieves S and T on c such that $T \in J(c)$ and $S \supseteq T$, $S \in J(c)$;
- (c) if $R \in J(c)$ then for any arrow $g : d \rightarrow c$ there exists a sieve $S \in J(d)$ such that for each arrow f in S , $g \circ f \in R$;
- (d) if the sieve S generated by a presieve $\{f_i : c_i \rightarrow c \mid i \in I\}$ belongs to $J(c)$ and for each $i \in I$ we have a presieve $\{g_{ij} : d_{ij} \rightarrow c_i \mid j \in I_i\}$ such that the sieve T_i generated by it belongs to $J(c_i)$, then the sieve R generated by the family of composites $\{f_i \circ g_{ij} : d_{ij} \rightarrow c \mid i \in I, j \in I_i\}$ belongs to $J(c)$.

In point (d) of this definition, the sieve R will be called the *composite* of the sieve S with the sieves T_i for $i \in I$ and denoted by $S * \{T_i \mid i \in I\}$.

Let us prove the equivalence of the two definitions.

First, let us assume the classical definition and derive the second. To prove property (b), let us assume that $S \supseteq T$ with $T \in J(c)$; then for every arrow f in T we have $f^*(S) \supseteq f^*(T) = M_c \in J(c)$ so by the transitivity axiom $S \in J(c)$, as required. Property (c) immediately follows from the stability axiom. Property (d) follows from the transitivity axiom by observing that for all arrows f in S , $f^*(R)$ is J -covering. Indeed, if $f \in S$ then $f = f_i \circ h$ for some $i \in I$ and arrow h ; so $f^*(R) = h^*(f_i^*(R)) \supseteq h^*(T_i) \in J(\text{dom}(h))$ and hence $f^*(R) \in J(\text{dom}(f))$ by property (b) and the stability axiom.

Conversely, let us assume the second definition and derive the classical one. The stability axiom easily follows from (b) and (c); indeed, if $R \in J(c)$ and $g : d \rightarrow c$ is an arrow with codomain c , then $h^*(R)$ contains the sieve S given by property (c) and hence it is J -covering by property (b). To prove the transitivity axiom it suffices to observe that, given a sieve R on c and a sieve $S \in J(c)$ such that $h^*(R)$ is J -covering for all arrows h in S , R contains the composite of the sieve S with the sieves of the form $h^*(R)$ for h in S .

Note that, in Definition 3.1.3, one can equivalently require in property (d) that the presieves $\{f_i : c_i \rightarrow c \mid i \in I\}$ and $\{g_{ij} : d_{ij} \rightarrow c_i \mid j \in I_i\}$ be sieves; indeed, it is clear from the above arguments that both versions of the condition are equivalent, assuming properties (a), (b) and (c), to the transitivity axiom.

In the case when the category $\mathcal{C}$ has pullbacks, property (c) (equivalently, the stability axiom) may be replaced by the following condition: if the sieve generated by $\{f_i : c_i \rightarrow c \mid i \in I\}$ belongs to $J(c)$ then, for any arrow $g : d \rightarrow c$, the sieve generated by the family obtained by pulling back the arrows f_i along g belongs to $J(d)$.

Remark 3.1.4 The operation of composition of sieves in $\mathcal{C}$ as in Definition 3.1.3(d) behaves naturally with respect to the operator $\overline{(-)}^J$ of J -closure of sieves for a Grothendieck topology J on $\mathcal{C}$; that is, with the above notation, we have

$$\overline{S * \{T_i \mid i \in I\}}^J = \overline{S * \{\overline{T_i}^J \mid i \in I\}}^J.$$

To verify this equality, it clearly suffices to prove that $S * \{\overline{T_i}^J \mid i \in I\} \subseteq \overline{S * \{T_i \mid i \in I\}}^J$; but this easily follows from property (b) in Definition 3.1.3.

3.1.3 Generators for Grothendieck topologies

Recall from section 1.4.2 that, if $\mathcal{C}$ is a regular category, we may define the regular topology $J_{\mathcal{C}}^{\text{reg}}$ as the Grothendieck topology on $\mathcal{C}$ having as sieves exactly those which contain a cover, while if $\mathcal{C}$ is a geometric category, we may define the geometric topology $J_{\mathcal{C}}^{\text{geom}}$ as the Grothendieck topology on $\mathcal{C}$ having as sieves exactly those which contain a small covering family.

The following result about these topologies holds. Below, by a *principal sieve* we mean a sieve which is generated by a single arrow; we denote by (f) the principal sieve generated by an arrow f .

Proposition 3.1.5 *Let $\mathcal{C}$ be a category and J a Grothendieck topology on it.*

- (i) *If $\mathcal{C}$ is regular and $J \supseteq J_{\mathcal{C}}^{\text{reg}}$ then J is generated over $J_{\mathcal{C}}^{\text{reg}}$ by a collection of sieves generated by a family of monomorphisms.*
- (ii) *If $\mathcal{C}$ is geometric and $J \supseteq J_{\mathcal{C}}^{\text{geom}}$ then J is generated over $J_{\mathcal{C}}^{\text{geom}}$ by a collection of principal sieves generated by single monomorphisms.*

Proof

- (i) Given an object $c \in \mathcal{C}$ and a sieve R on c in $\mathcal{C}$, let us denote, for each arrow r in R , by $\text{dom}(r) \xrightarrow{r''} x \xrightarrow{r'} c$ its cover-mono factorization in $\mathcal{C}$ and by R' the sieve in $\mathcal{C}$ generated by the arrows r' (for r in R). Clearly, it is enough to prove that $R \in J(c)$ if and only if $R' \in J(c)$. The ‘only if’ part follows from property (b) in Definition 3.1.3 since $R' \supseteq R$, while the ‘if’ part follows from property (d) in Definition 3.1.3 by noticing that, since $J \supseteq J_{\mathcal{C}}^{\text{reg}}$, all the principal sieves generated by the arrows r'' (for $r \in R$) are J -covering.
- (ii) Given an object $c \in \mathcal{C}$ and a sieve R on $c \in \mathcal{C}$, let r be the subobject of c given by the union in $\text{Sub}_{\mathcal{C}}(c)$ of all the images in $\mathcal{C}$ of the morphisms in R (this union exists because, $\mathcal{C}$ being well-powered, there is, up to isomorphism, only a set of monomorphisms with a given codomain).

Clearly, it is enough to prove that $R \in J(c)$ if and only if $(r) \in J(c)$. The ‘only if’ part follows from property (b) in Definition 3.1.3 since $(r) \supseteq R$, while the ‘if’ part follows from property (d) in Definition 3.1.3 by observing that, since $J \supseteq J_{\mathcal{C}}^{\text{geom}}$, the sieve generated by the inclusions into r of the images of the morphisms in R is J -covering. $\square$

Given a sieve R on a regular category $\mathcal{C}$, it is natural to consider the sieve R^{reg} generated by all the images of morphisms in R ; similarly, if $\mathcal{C}$ is a geometric category, it is natural to consider the sieve R^{geom} generated by the union (in the appropriate subobject lattice) of all the images of morphisms in R . The following result provides a link between these constructions and the notions of regular and geometric topology:

Proposition 3.1.6 *Let R be a sieve on a category $\mathcal{C}$.*

- (i) *If $\mathcal{C}$ is a regular category then $R^{\text{reg}} = \overline{R}^{J_{\mathcal{C}}^{\text{reg}}}$.*
- (ii) *If $\mathcal{C}$ is a geometric category then $R^{\text{geom}} = \overline{R}^{J_{\mathcal{C}}^{\text{geom}}}$.*

Proof

- (i) Let us begin by proving that the inclusion $R^{\text{reg}} \subseteq \overline{R}^{J_{\mathcal{C}}^{\text{reg}}}$ holds. Clearly, it suffices to show that for any f in R , $\text{Im}(f) \in \overline{R}^{J_{\mathcal{C}}^{\text{reg}}}$. Now, if $a \xrightarrow{f'} a' \xrightarrow{\text{Im}(f)} b$ is the cover-mono factorization of f then $(f') \subseteq \text{Im}(f)^*(R)$; but f' is a cover, so $(f') \in J_{\mathcal{C}}^{\text{reg}}(a')$ and hence $\text{Im}(f)^*(R)$ is $J_{\mathcal{C}}^{\text{reg}}$ -covering by property (b) in Definition 3.1.3. It remains to prove the converse inclusion. If $f \in \overline{R}^{J_{\mathcal{C}}^{\text{reg}}}$ then $f^*(R)$ contains a cover, call it h . Since composition of covers is a cover then f factors through $\text{Im}(f \circ h)$ and hence $f \in R^{\text{reg}}$, as required.
- (ii) Let R be a sieve generated by a presieve $\{r_i \mid i \in I\}$ on an object $c \in \mathcal{C}$ (for our purposes we can suppose I to be a set without loss of generality, every geometric category being well-powered). Let us denote by $r : d \rightarrow c$ the union in $\text{Sub}_{\mathcal{C}}(c)$ of the subobjects $\text{Im}(r_i)$ as i varies in I and by h_i the (unique) factorization of r_i through r (for each $i \in I$). To prove the inclusion $R^{\text{geom}} \subseteq \overline{R}^{J_{\mathcal{C}}^{\text{geom}}}$, it is enough to show that $r \in \overline{R}^{J_{\mathcal{C}}^{\text{geom}}}$. Now, $r = \bigcup_{i \in I} \text{Im}(r_i)$ so $1_d = r^*(\bigcup_{i \in I} \text{Im}(r_i)) = \bigcup_{i \in I} r^*(\text{Im}(r \circ h_i)) = \bigcup_{i \in I} \text{Im}(r^*(r \circ h_i)) = \bigcup_{i \in I} \text{Im}(h_i)$, where the second and third equalities follow from the fact that in any geometric category cover-mono factorizations and small unions of subobjects are stable under pullback and the last equality follows from the fact that r is monic. So $\{h_i \mid i \in I\}$ is a small covering family contained in $r^*(R)$ and hence $r^*(R)$ is $J_{\mathcal{C}}^{\text{geom}}$ -covering, as required. Conversely, let us suppose that, given $f : d \rightarrow c$, $f^*(R)$ contains a small covering family $\{h_j \mid j \in J\}$. We want to prove that f factors

through r . Since r is monic, this condition is clearly equivalent to requiring that $f^*(r) = 1_d$. Now, $f^*(r) = f^*(\bigcup_{i \in I} \text{Im}(r_i)) = \bigcup_{i \in I} \text{Im}(f^*(r_i))$. For each $j \in J$ there exists $i \in I$ such that $f \circ h_j \in (r_i)$ and hence h_j factors through $f^*(r_i)$; this in turn implies that $\text{Im}(h_j)$ factors through $\text{Im}(f^*(r_i))$, whence $\bigcup_{i \in I} \text{Im}(f^*(r_i)) \supseteq \bigcup_{j \in J} \text{Im}(h_j) = 1_d$. Therefore $f^*(r) = 1_d$, as required. $\square$

Remark 3.1.7 It follows as an immediate consequence of Proposition 3.1.6 that if $\mathcal{C}$ is a regular (resp. geometric) category then, for any sieve R on c and any arrow $f : d \rightarrow c$, $f^*(R^{\text{reg}}) = f^*(R)^{\text{reg}}$ (resp. $f^*(R^{\text{geom}}) = f^*(R)^{\text{geom}}$); indeed, universal closure operations always commute with pullbacks.

3.2 Quotients and subtoposes

3.2.1 The duality theorem

In this section, we establish a duality theorem providing a bijection between the subtoposes of the classifying topos of a given geometric theory $\mathbb{T}$ and the ‘quotients’ of $\mathbb{T}$ (considered up to a syntactic notion of equivalence).

Let us start with an easy remark: every subtopos (in the Lawvere-Tierney sense) of a Grothendieck topos is a Grothendieck topos. This can be seen in (at least) two different ways, as follows.

Recall that a subtopos of a topos $\mathcal{E}$ is a geometric inclusion of the form $\mathbf{sh}_j(\mathcal{E}) \hookrightarrow \mathcal{E}$ for a local operator j on $\mathcal{E}$, equivalently an equivalence class of geometric inclusions to the topos $\mathcal{E}$. By Theorem 1.3.36, the subtoposes of a presheaf topos $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ are in bijection with the Grothendieck topologies J on the category $\mathcal{C}$, i.e. every geometric inclusion to $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ is, up to equivalence, of the canonical form $\mathbf{Sh}(\mathcal{C}, J) \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ for a unique Grothendieck topology J on $\mathcal{C}$; moreover, such a geometric inclusion $\mathbf{Sh}(\mathcal{C}, J) \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ factors through another canonical geometric inclusion $\mathbf{Sh}(\mathcal{C}, J') \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ if and only if $J' \subseteq J$ (i.e. every J' -covering sieve is a J -covering sieve). Now, the geometric inclusions to a Grothendieck topos $\mathbf{Sh}(\mathcal{C}, J)$ can be clearly identified with the geometric inclusions to $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ which factor through $\mathbf{Sh}(\mathcal{C}, J) \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ and hence the subtoposes of $\mathbf{Sh}(\mathcal{C}, J)$ correspond precisely to the Grothendieck topologies J' on $\mathcal{C}$ such that $J' \supseteq J$. This provides us with the first proof of our remark. Alternatively, we can argue as follows. By Theorem C2.2.8 [55], an elementary topos $\mathcal{E}$ is a Grothendieck topos if and only if there exists a bounded geometric morphism $\mathcal{E} \rightarrow \mathbf{Set}$ (in the sense of Definition B3.1.7 [55]). Now, a geometric inclusion is always a localic morphism (cf. Example A4.6.2(a) [55]), and hence a bounded morphism (cf. Example B3.1.8 [50]); but a composite of bounded morphism is a bounded morphism (by Lemma B3.1.10(i) [55]), whence our claim follows immediately.

This remark is fundamental for our purposes for the following reason. For each elementary topos $\mathcal{E}$, the collection of subtoposes of $\mathcal{E}$ has the structure of a

coHeyting algebra (cf. Example A4.5.13(f) [55]), and there are many important concepts in topos theory that apply in this context (cf. section A4 of [50]); it is therefore natural to investigate their meaning in the setting of Grothendieck toposes. In fact, thanks to the duality theorem established below, we will also be able to interpret all these concepts in the context of geometric theories (cf. Chapter 4).

Before we can state our duality theorem, which describes how the relationship between Grothendieck toposes and geometric theories given by the theory of classifying toposes ‘restricts’ to the context of subtoposes of a given Grothendieck topos, we need to introduce some definitions. Concerning terminology, we use the term *theory* in the sense of axiomatic presentation (as in Definition 1.2.5); accordingly, we consider two theories over a given signature equal when they have exactly the same axioms.

Definition 3.2.1 Let $\mathbb{T}$ be a geometric theory over a signature Σ and σ, σ' two geometric sequents over Σ . Then σ and σ' are said to be $\mathbb{T}$ -*equivalent* if σ is provable in $\mathbb{T} \cup \{\sigma'\}$ and σ' is provable in $\mathbb{T} \cup \{\sigma\}$.

Definition 3.2.2 Let $\mathbb{T}$ be a geometric theory over a signature Σ . A *quotient* of $\mathbb{T}$ is a geometric theory $\mathbb{T}'$ over Σ such that every axiom of $\mathbb{T}$ is provable in $\mathbb{T}'$.

Remark 3.2.3 The notion of provability in geometric logic to which we refer here is the one defined in section 1.2.3; that system is essentially constructive, but, by Theorem 1.2.13, we may add the Law of Excluded Middle to it (thus making it classical) without affecting the corresponding notion of provability.

Definition 3.2.4 Let $\mathbb{T}$ and $\mathbb{T}'$ be geometric theories over a signature Σ . We say that $\mathbb{T}$ and $\mathbb{T}'$ are *syntactically equivalent*, and we write $\mathbb{T} \equiv_s \mathbb{T}'$, if for every geometric sequent σ over Σ , σ is provable in $\mathbb{T}$ if and only if σ is provable in $\mathbb{T}'$.

We are now ready to state the duality theorem between quotients and subtoposes.

Theorem 3.2.5 *Let $\mathbb{T}$ be a geometric theory over a signature Σ . Then the assignment sending a quotient of $\mathbb{T}$ to its classifying topos yields a bijection between the $\equiv_s$ -equivalence classes of quotients of $\mathbb{T}$ and the subtoposes of the classifying topos $\mathbf{Set}[\mathbb{T}]$ of $\mathbb{T}$.*

Proof Recall from section 2.1.2 that the classifying topos $\mathbf{Set}[\mathbb{T}]$ of $\mathbb{T}$ can be represented as $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$, where $\mathcal{C}_{\mathbb{T}}$ is the geometric syntactic category of $\mathbb{T}$ and $J_{\mathbb{T}}$ is the canonical topology on it (i.e. the Grothendieck topology on $\mathcal{C}_{\mathbb{T}}$ having as covering sieves exactly those which contain small covering families) and that we have an equivalence of categories $\mathbb{T}\text{-mod}(\mathcal{E}) \simeq \mathbf{Flat}_{J_{\mathbb{T}}}(\mathcal{C}_{\mathbb{T}}, \mathcal{E})$ (natural in $\mathcal{E} \in \mathbf{BTop}$) which sends each model $M \in \mathbb{T}\text{-mod}(\mathcal{E})$ to the functor $F_M : \mathcal{C}_{\mathbb{T}} \rightarrow \mathcal{E}$ assigning to a formula $\{\vec{x}. \phi\}$ (the domain of) its interpretation $[[\vec{x}. \phi]]_M$ in M .

Notice that $\mathcal{C}_{\mathbb{T}}$ is an essentially small category, i.e. it is equivalent to a small category (cf. the proof of Lemma D1.4.10(iv) [55]); hence all the results valid for small sites naturally extend to sites with underlying category $\mathcal{C}_{\mathbb{T}}$.

Let us recall the construction of pullbacks in $\mathcal{C}_{\mathbb{T}}$. Given two morphisms

$$\{\vec{x} . \phi\} \xrightarrow{[\theta]} \{\vec{y} . \psi\}$$

and

$$\{\vec{x}' . \phi'\} \xrightarrow{[\theta']} \{\vec{y} . \psi\}$$

in $\mathcal{C}_{\mathbb{T}}$ with common codomain, we have the following pullback square in $\mathcal{C}_{\mathbb{T}}$:

$$\begin{array}{ccc} \{\vec{x}, \vec{x}' . (\exists \vec{y})(\theta[\vec{x}/\vec{x}] \wedge \theta'[\vec{x}'/\vec{x}'])\} & \xrightarrow{[(\exists \vec{y})(\theta \wedge \theta' \wedge \vec{x} = \vec{x}')] } & \{\vec{x}' . \phi'\} \\ \downarrow [(\exists \vec{y})(\theta \wedge \theta' \wedge \vec{x} = \vec{x})] & & \downarrow [\theta'] \\ \{\vec{x} . \phi\} & \xrightarrow{[\theta]} & \{\vec{y} . \psi\}. \end{array}$$

We note that the sequent $(\phi' \vdash_{\vec{x}'} (\exists \vec{x}, \vec{x}')((\exists \vec{y})(\theta \wedge \theta' \wedge \vec{x} = \vec{x}')))$ is provable in $\mathbb{T}$ from the sequent $(\psi \vdash_{\vec{y}} (\exists \vec{x})\theta)$. Indeed, it is clearly equivalent in geometric logic to the sequent $(\phi' \vdash_{\vec{x}'} (\exists \vec{y})(\theta \wedge \theta'))$, and the sequents $(\phi' \vdash_{\vec{x}'} (\exists \vec{y})\theta')$ and $(\theta' \vdash_{\vec{x}', \vec{y}} \psi)$ are provable in $\mathbb{T}$.

Next, we observe that, given a $\mathbb{T}$ -model M in a Grothendieck topos $\mathcal{E}$, $F_M : \mathcal{C}_{\mathbb{T}} \rightarrow \mathcal{E}$ sends a small family $\{\theta_i \mid i \in I\}$ of morphisms

$$\{\vec{x}_i . \phi_i\} \xrightarrow{[\theta_i]} \{\vec{y} . \psi\}$$

in $\mathcal{C}_{\mathbb{T}}$ with common codomain to an epimorphic family in $\mathcal{E}$ if and only if $[[\vec{y} . \psi]]_M = [[\vec{y} . \bigvee_{i \in I} (\exists \vec{x}_i)\theta_i]]_M$, equivalently if and only if the sequent $(\psi \vdash_{\vec{y}} \bigvee_{i \in I} (\exists \vec{x}_i)\theta_i)$ is satisfied in M .

These remarks show, by the soundness theorem for geometric logic (cf. Theorem 1.4.15), that for any small presieve R in $\mathcal{C}_{\mathbb{T}}$, the $J_{\mathbb{T}}$ -continuous flat functors on $\mathcal{C}_{\mathbb{T}}$ sending R to an epimorphic family also send all the pullbacks of R along arrows in $\mathcal{C}_{\mathbb{T}}$ to epimorphic families. It thus follows from Remark 3.1.2 and Lemma 3 [7] that the $J_{\mathbb{T}}$ -continuous flat functors on $\mathcal{C}_{\mathbb{T}}$ which send each of the small presieves in a given collection $\mathcal{F}$ to an epimorphic family coincide with the $J_{\mathbb{T}}$ -continuous flat functors on $\mathcal{C}_{\mathbb{T}}$ which are $J_{\mathcal{F}}$ -continuous, where $J_{\mathcal{F}}$ is the Grothendieck topology on $\mathcal{C}_{\mathbb{T}}$ generated over $J_{\mathbb{T}}$ by the sieves generated by the presieves in $\mathcal{F}$.

Given a quotient $\mathbb{T}'$ of $\mathbb{T}$, we may construct its classifying topos as follows. Let $\mathbb{T}'$ be obtained from $\mathbb{T}$ by adding a number of geometric sequents over Σ ; of course, up to syntactic equivalence, there are many possible ways of presenting $\mathbb{T}'$ in such a form (for example one may take as axioms all the axioms of $\mathbb{T}'$ or, more economically, all the axioms of $\mathbb{T}'$ which are not provable in $\mathbb{T}$), but our

construction will be independent from any particular presentation. For each of these axioms $(\phi \vdash_{\vec{x}} \psi)$, consider the corresponding monomorphism

$$\{\vec{x}' \cdot \phi \wedge \psi\} \xrightarrow{[(\phi \wedge \psi \wedge \vec{x}' = \vec{x})]} \{\vec{x} \cdot \phi\}$$

in the geometric syntactic category $\mathcal{C}_{\mathbb{T}}$ of $\mathbb{T}$.

It is clear that, given a $\mathbb{T}$ -model M in a Grothendieck topos $\mathcal{E}$, the functor $F_M : \mathcal{C}_{\mathbb{T}} \rightarrow \mathcal{E}$ sends the morphism

$$\{\vec{x}' \cdot \phi \wedge \psi\} \xrightarrow{[(\phi \wedge \psi \wedge \vec{x}' = \vec{x})]} \{\vec{x} \cdot \phi\}$$

to an epimorphism if and only if $[[\vec{x} \cdot \phi]]_M \leq [[\vec{x} \cdot \psi]]_M$ i.e. if and only if the sequent $(\phi \vdash_{\vec{x}} \psi)$ holds in M .

So the $J_{\mathbb{T}}$ -continuous flat functors on $\mathcal{C}_{\mathbb{T}}$ which send each of the monomorphisms corresponding to the axioms of $\mathbb{T}'$ to an epimorphism correspond to the models of $\mathbb{T}'$. Therefore, if $J_{\mathbb{T}'}^{\mathbb{T}}$ is the smallest Grothendieck topology on $\mathcal{C}_{\mathbb{T}}$ for which all the $J_{\mathbb{T}}$ -covering sieves and the sieves containing a morphism corresponding to an axiom of $\mathbb{T}'$ are covering then, by Diaconescu's equivalence, the topos $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}'}^{\mathbb{T}})$ classifies the theory $\mathbb{T}'$, and the canonical geometric inclusion $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}'}^{\mathbb{T}}) \hookrightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}^{\mathbb{T}})$ corresponding to the inclusion $J_{\mathbb{T}} \subseteq J_{\mathbb{T}'}^{\mathbb{T}}$, realizes $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}'}^{\mathbb{T}})$ as a subtopos of $\mathbf{Set}[\mathbb{T}]$.

Now, to have a well-defined assignment from the $\equiv_s$ -equivalence classes of quotients of $\mathbb{T}$ to the subtoposes of $\mathbf{Set}[\mathbb{T}]$, it remains to verify that the topology $J_{\mathbb{T}'}^{\mathbb{T}}$ does not depend on the particular choice of axioms for $\mathbb{T}'$, i.e. that it is the same for all the quotients in a given $\equiv_s$ -equivalence class.

Let $\mathbb{T}_1$ and $\mathbb{T}_2$ be quotients of $\mathbb{T}$ such that $\mathbb{T}_1 \equiv_s \mathbb{T}_2$; we want to prove that $J_{\mathbb{T}_1}^{\mathbb{T}} = J_{\mathbb{T}_2}^{\mathbb{T}}$. We will prove the existence of a geometric equivalence $\tau : \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}_1}^{\mathbb{T}}) \rightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}_2}^{\mathbb{T}})$ such that the diagram in $\mathfrak{B}\mathfrak{T}\mathfrak{o}\mathfrak{p}$

$$\begin{array}{ccc} \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}_1}^{\mathbb{T}}) & \xrightarrow{\tau} & \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}_2}^{\mathbb{T}}) \\ & \searrow i_1 \quad \swarrow i_2 & \\ & [\mathcal{C}_{\mathbb{T}}^{\text{op}}, \mathbf{Set}] & \end{array}$$

where the geometric inclusions

$$\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}_1}^{\mathbb{T}}) \rightarrow [(\mathcal{C}_{\mathbb{T}})^{\text{op}}, \mathbf{Set}]$$

and

$$\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}_2}^{\mathbb{T}}) \rightarrow [(\mathcal{C}_{\mathbb{T}})^{\text{op}}, \mathbf{Set}]$$

are the canonical ones, commutes up to isomorphism.

From the identification of equivalence classes of geometric inclusions to a given topos with local operators on that topos (cf. section 1.3.4) the equality of

the two topologies $J_{\mathbb{T}_1}^{\mathbb{T}}$ and $J_{\mathbb{T}_2}^{\mathbb{T}}$ will then follow. By the 2-dimensional Yoneda lemma recalled in the section 3.1.1, it is equivalent to show the existence of an equivalence of categories

$$l_{\mathcal{E}} : \mathbf{Geom}(\mathcal{E}, \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}_1}^{\mathbb{T}})) \rightarrow \mathbf{Geom}(\mathcal{E}, \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}_2}^{\mathbb{T}}))$$

natural in $\mathcal{E} \in \mathfrak{BTop}$ such that $(i_1 \circ -) \circ l_{\mathcal{E}} \cong (i_2 \circ -)$ for each $\mathcal{E} \in \mathfrak{BTop}$. Since $\mathbb{T}_1 \equiv_s \mathbb{T}_2$, $\mathbb{T}_1$ and $\mathbb{T}_2$ have the same models (in every Grothendieck topos) and hence we may obtain such an equivalence by composing

$$\mathbf{Geom}(\mathcal{E}, \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}_1}^{\mathbb{T}})) \simeq \mathbf{Flat}_{J_{\mathbb{T}_1}}(\mathcal{C}_{\mathbb{T}}, \mathcal{E}) \simeq \mathbb{T}_1\text{-mod}(\mathcal{E}) = \mathbb{T}_2\text{-mod}(\mathcal{E})$$

with

$$\mathbb{T}_2\text{-mod}(\mathcal{E}) \simeq \mathbf{Flat}_{J_{\mathbb{T}_2}}(\mathcal{C}_{\mathbb{T}}, \mathcal{E}) \simeq \mathbf{Geom}(\mathcal{E}, \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}_2}^{\mathbb{T}})),$$

where the first and last equivalences are instances of Diaconescu's equivalence.

Conversely, let $\mathcal{E} \hookrightarrow \mathbf{Set}[\mathbb{T}]$ be a subtopos of $\mathbf{Set}[\mathbb{T}]$; up to equivalence, this subtopos has the form $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J) \hookrightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ for a unique Grothendieck topology J such that $J \supseteq J_{\mathbb{T}}$. Let us prove that there exists a quotient $\mathbb{T}^J$ of $\mathbb{T}$ classified by it. Let $\mathbb{T}^J$ be the theory consisting of all (the axioms of $\mathbb{T}$ and) the geometric sequents over Σ of the form $(\psi \vdash_{\vec{y}} (\exists \vec{x})\theta)$, where $[\theta]$ is any monomorphism

$$\{\vec{x} \cdot \phi\} \xrightarrow{[\theta]} \{\vec{y} \cdot \psi\}$$

in $\mathcal{C}_{\mathbb{T}}$ generating a J -covering sieve.

Since, for any $\mathbb{T}$ -model M in a Grothendieck topos $\mathcal{E}$, the functor F_M sends $[\theta]$ to an epimorphism if and only if the sequent $(\psi \vdash_{\vec{y}} (\exists \vec{x})\theta)$ holds in M , it follows from Remark 3.1.2, Proposition 3.1.5 and Lemma 3 [7] that the equivalence $\mathbb{T}\text{-mod}(\mathcal{E}) \simeq \mathbf{Flat}_{J_{\mathbb{T}}}(\mathcal{C}_{\mathbb{T}}, \mathcal{E})$ restricts to an equivalence $\mathbb{T}^J\text{-mod}(\mathcal{E}) \simeq \mathbf{Flat}_J(\mathcal{C}_{\mathbb{T}}, \mathcal{E})$ (naturally in $\mathcal{E} \in \mathfrak{BTop}$) and hence that $\mathcal{E} = \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J)$ classifies the theory $\mathbb{T}^J$.

To conclude the proof of the theorem it remains to show that the two assignments $\mathbb{T}' \rightarrow J_{\mathbb{T}'}^{\mathbb{T}}$ and $J \rightarrow \mathbb{T}^J$ are bijections inverse to each other between the $\equiv_s$ -equivalence classes of quotients of $\mathbb{T}$ and the subtoposes of the classifying topos $\mathbf{Set}[\mathbb{T}]$ of $\mathbb{T}$.

To prove that for any quotient $\mathbb{T}'$ of $\mathbb{T}$ we have $\mathbb{T}' \equiv_s \mathbb{T}^{J_{\mathbb{T}'}}_{\mathbb{T}}$, we argue as follows. First, we observe that for any $\mathbb{T}$ -model M in a Grothendieck topos $\mathcal{E}$, M is a $\mathbb{T}'$ -model if and only if it is a $\mathbb{T}^{J_{\mathbb{T}'}}_{\mathbb{T}}$ -model; indeed, by definition of $J_{\mathbb{T}'}$ and of $\mathbb{T}^{J_{\mathbb{T}'}}_{\mathbb{T}}$, both the $\mathbb{T}'$ -models and the $\mathbb{T}^{J_{\mathbb{T}'}}_{\mathbb{T}}$ -models in $\mathcal{E}$ correspond to the functors in $\mathbf{Flat}_{J_{\mathbb{T}'}}(\mathcal{C}_{\mathbb{T}'}, \mathcal{E})$ via the equivalence $\mathbf{Flat}_{J_{\mathbb{T}'}}(\mathcal{C}_{\mathbb{T}'}, \mathcal{E}) \simeq \mathbb{T}'\text{-mod}(\mathcal{E})$.

Now, consider the image $U_{\mathbb{T}'}^{\mathbb{T}}$ of $a_{J_{\mathbb{T}'}} \circ y^{\mathbb{T}}$ in $\mathbb{T}'\text{-mod}(\mathcal{E})$ through the equivalence $\mathbf{Flat}_{J_{\mathbb{T}'}}(\mathcal{C}_{\mathbb{T}'}, \mathcal{E}) \simeq \mathbb{T}'\text{-mod}(\mathcal{E})$, where $y^{\mathbb{T}} : \mathcal{C}_{\mathbb{T}} \rightarrow [(\mathcal{C}_{\mathbb{T}})^{\text{op}}, \mathbf{Set}]$ is the Yoneda embedding and $a_{J_{\mathbb{T}'}} : [(\mathcal{C}_{\mathbb{T}})^{\text{op}}, \mathbf{Set}] \rightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}'}^{\mathbb{T}})$ is the associated sheaf functor. By Diaconescu's equivalence and the naturality in $\mathcal{E} \in \mathfrak{BTop}$ of the equivalences

$$\mathbb{T}'\text{-mod}(\mathcal{E}) = \mathbb{T}^{J_{\mathbb{T}'}}_{\mathbb{T}'}\text{-mod}(\mathcal{E}) \simeq \mathbf{Flat}_{J_{\mathbb{T}'}}(\mathcal{C}_{\mathbb{T}'}, \mathcal{E}) \simeq \mathbf{Geom}(\mathcal{E}, \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}'}^{\mathbb{T}})),$$

the Σ -structure $U_{\mathbb{T}'}^{\mathbb{T}}$ is a universal model for both $\mathbb{T}'$ and $\mathbb{T}^{J_{\mathbb{T}'}}_{\mathbb{T}}$; in particular, it is conservative both as a $\mathbb{T}'$ -model and as a $\mathbb{T}^{J_{\mathbb{T}'}}_{\mathbb{T}}$ -model (cf. Remark 2.1.9). From this it clearly follows that $\mathbb{T}' \equiv_s \mathbb{T}^{J_{\mathbb{T}'}}_{\mathbb{T}}$, as required.

On the other hand, the fact that $J = J_{\mathbb{T}, \mathbb{T}'}$ directly follows from the definition of the assignment $\mathbb{T}' \rightarrow J_{\mathbb{T}, \mathbb{T}'}$. $\square$

We shall refer to the topology $J_{\mathbb{T}, \mathbb{T}'}$ as the *associated $\mathbb{T}$ -topology of $\mathbb{T}'$* and to the ($\equiv_s$ -equivalence class of the) quotient $\mathbb{T}^J$ as the *associated $\mathbb{T}$ -quotient of J* .

For each Grothendieck topos $\mathcal{E}$, we denote by

$$\tau^{\mathcal{E}} : \mathbb{T}\text{-mod}(\mathcal{E}) \simeq \mathbf{Geom}(\mathcal{E}, \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}))$$

the composite of the equivalence

$$\mathbb{T}\text{-mod}(\mathcal{E}) \simeq \mathbf{Flat}_{J_{\mathbb{T}}}(\mathcal{C}_{\mathbb{T}}, \mathcal{E})$$

considered in the proof of Theorem 3.2.5 with Diaconescu's equivalence

$$\mathbf{Flat}_{J_{\mathbb{T}}}(\mathcal{C}_{\mathbb{T}}, \mathcal{E}) \simeq \mathbf{Geom}(\mathcal{E}, \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})).$$

Given a quotient $\mathbb{T}'$ of a theory $\mathbb{T}$, we denote by $i_{\mathbb{T}'}^{\mathcal{E}} : \mathbb{T}'\text{-mod}(\mathcal{E}) \hookrightarrow \mathbb{T}\text{-mod}(\mathcal{E})$ the canonical inclusion functor.

Remark 3.2.6 With the above notation, given a quotient $\mathbb{T}'$ of $\mathbb{T}$ with associated $\mathbb{T}$ -topology J on $\mathcal{C}_{\mathbb{T}}$ then, denoting by i_J the canonical geometric inclusion $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J) \hookrightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$, the diagram in $\mathfrak{Cat}$

$$\begin{array}{ccc} \mathbb{T}'\text{-mod}(\mathcal{E}) & \xrightarrow{\simeq} & \mathbf{Geom}(\mathcal{E}, \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J)) \\ \downarrow i_{\mathbb{T}'}^{\mathcal{E}} & & \downarrow i_J \circ - \\ \mathbb{T}\text{-mod}(\mathcal{E}) & \xrightarrow[\tau^{\mathcal{E}}]{\simeq} & \mathbf{Geom}(\mathcal{E}, \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})) \end{array}$$

commutes (up to isomorphism) naturally in $\mathcal{E} \in \mathfrak{BTop}$.

It should be noted that our method of constructing the associated $\mathbb{T}$ -topology of a given quotient of $\mathbb{T}$ has points in common with the ‘forcing’ method expressed by Proposition D3.1.10 [55]; important precedents for this result are the paper [72] on ‘forcing topologies’ by M. Tierney and the work of M. and M.-F. Coste [39] on the classifying toposes of coherent theories (included in the Appendix of [64]). In fact, our arguments show that, more generally, it is always possible to construct the classifying topos of a quotient $\mathbb{T}'$ of a given cartesian (resp. regular, coherent, geometric) theory $\mathbb{T}$ as a category of sheaves on the cartesian (resp. regular, coherent, geometric) syntactic category of $\mathbb{T}$ starting from a way of expressing $\mathbb{T}'$ as a theory obtained from $\mathbb{T}$ by adding axioms of the form $(\psi \vdash_{\vec{y}} \bigvee_{i \in I} (\exists \vec{x}_i) \theta_i)$ where ψ and the θ_i are cartesian (resp. regular, coherent, geometric) formulae over the signature of $\mathbb{T}$ such that the θ_i are

$\mathbb{T}$ -provably functional formulae from a cartesian (resp. regular, coherent, geometric) formula-in-context to $\psi(\vec{y})$; see also Theorem 8.1.10, which arises from an implementation of the same idea in the case $\mathbb{T}$ is a theory of presheaf type, and which is an actual generalization of Proposition D3.1.10 [55].

Given a Grothendieck topos $\mathcal{E}$ and a signature Σ , it is natural to wonder under which conditions there exists a geometric theory $\mathbb{T}$ over Σ classified by $\mathcal{E}$. Theorem 3.2.5 gives an answer to this question: there exists such a theory if and only if there exists a geometric inclusion from $\mathcal{E}$ to the classifying topos of the empty theory over Σ .

3.2.2 The proof-theoretic interpretation

In this section we provide an alternative, syntactic, proof of Theorem 3.2.5. This will be based on a logical interpretation of the notion of Grothendieck topology. Specifically, given a collection $\mathcal{A}$ of sieves on a given category $\mathcal{C}$, the notion of Grothendieck topology on $\mathcal{C}$ naturally gives rise to a proof system $\mathcal{T}_{\mathcal{C}}^{\mathcal{A}}$, as follows: the axioms of $\mathcal{T}_{\mathcal{C}}^{\mathcal{A}}$ are the sieves in $\mathcal{A}$ plus all the maximal sieves, while the inference rules of $\mathcal{T}_{\mathcal{C}}^{\mathcal{A}}$ are the proof-theoretic versions of the well-known axioms for Grothendieck topologies, i.e.

Stability rule:

$$\frac{R}{f^*(R)}$$

where R is any sieve on an object c of $\mathcal{C}$ and f is any arrow in $\mathcal{C}$ with codomain c .

Transitivity rule:

$$\frac{Z \quad \{f^*(R) \mid f \in Z\}}{R}$$

where R and Z are sieves in $\mathcal{C}$ on a given object of $\mathcal{C}$.

Notice that the ‘closed theories’ of this proof system are precisely the Grothendieck topologies on $\mathcal{C}$ which contain the sieves in $\mathcal{A}$ as covering sieves and that the closure of a ‘theory’ in $\mathcal{T}_{\mathcal{C}}^{\mathcal{A}}$, i.e. of a collection $\mathcal{U}$ of sieves in $\mathcal{C}$, is exactly the Grothendieck topology on $\mathcal{C}$ generated by $\mathcal{A}$ and $\mathcal{U}$.

The following theorem will give a ‘proof-theoretic equivalence’ between the deduction system of geometric logic over a given geometric theory $\mathbb{T}$ (as described in section 1.2.2) and the system $\mathcal{T}_{\mathcal{C}_{\mathbb{T}}}^{J_{\mathbb{T}}}$.

Given a geometric theory $\mathbb{T}$ over a signature Σ , let $\mathcal{S}$ be the collection of geometric sequents over Σ , $\hat{\mathcal{S}}$ the quotient of $\mathcal{S}$ by the relation of $\mathbb{T}$ -equivalence and $S(\mathcal{C}_{\mathbb{T}})$ the collection of sieves in the geometric syntactic category $\mathcal{C}_{\mathbb{T}}$ of $\mathbb{T}$.

Motivated by the proof of the duality theorem in section 3.2.1, we define two correspondences $\mathcal{F} : \mathcal{S} \rightarrow S(\mathcal{C}_{\mathbb{T}})$ and $\mathcal{G} : S(\mathcal{C}_{\mathbb{T}}) \rightarrow \hat{\mathcal{S}}$, as follows.

Given a geometric sequent $\sigma \equiv (\phi \vdash_{\vec{x}} \psi)$ over Σ , we set $\mathcal{F}(\sigma)$ equal to the principal sieve in $\mathcal{C}_{\mathbb{T}}$ generated by the monomorphism

$$\{\vec{x}' \cdot \phi \wedge \psi\} \xrightarrow{[(\phi \wedge \psi \wedge \vec{x}' = \vec{x})]} \{\vec{x} \cdot \phi\}.$$

Conversely, given a sieve R in $\mathcal{C}_{\mathbb{T}}$, we set $\mathcal{G}(R)$ equal to the $\mathbb{T}$ -equivalence class of any geometric sequent $(\psi \vdash_{\vec{y}} (\exists \vec{x})\theta)$ such that $[\theta]$ is a monomorphism

$$\{\vec{x} . \phi\} \xrightarrow{[\theta]} \{\vec{y} . \psi\}$$

in $\mathcal{C}_{\mathbb{T}}$ generating the principal sieve $\overline{R}^{J_{\mathbb{T}}}$ (cf. Proposition 3.1.6).

Applying the covariant powerset image functor $\mathcal{P}$ to $\mathcal{F}$ and $\mathcal{G}$, we obtain maps of posets

$$\mathcal{P}(\mathcal{F}) : \mathcal{P}(\mathcal{S}) \rightarrow \mathcal{P}(S(\mathcal{C}_{\mathbb{T}}))$$

and

$$\mathcal{P}(\mathcal{G}) : \mathcal{P}(S(\mathcal{C}_{\mathbb{T}})) \rightarrow \mathcal{P}(\tilde{\mathcal{S}}).$$

From now on we will write $\mathcal{F}(U)$ for $\mathcal{P}(\mathcal{F})(U)$ and $\mathcal{G}(V)$ for $\mathcal{P}(\mathcal{G})(V)$.

We have closure operators

$$\overline{(-)}^{\mathbb{T}} : \mathcal{P}(\mathcal{S}) \rightarrow \mathcal{P}(\mathcal{S})$$

and

$$\overline{(-)}^T : \mathcal{P}(S(\mathcal{C}_{\mathbb{T}})) \rightarrow \mathcal{P}(S(\mathcal{C}_{\mathbb{T}}))$$

defined as follows: for a collection U of geometric sequents over Σ , $\overline{U}^{\mathbb{T}}$ is the collection of geometric sequents σ which are provable in $\mathbb{T} \cup U$ using geometric logic, while, for a collection V of sieves in $\mathcal{C}_{\mathbb{T}}$, $\overline{V}^T$ is the Grothendieck topology in $\mathcal{C}_{\mathbb{T}}$ generated by $J_{\mathbb{T}}$ and V (i.e. the smallest Grothendieck topology J on $\mathcal{C}_{\mathbb{T}}$ such that all the $J_{\mathbb{T}}$ -covering sieves and the sieves in V are J -covering). Note that the relation of $\mathbb{T}$ -equivalence on $\mathcal{S}$ is compatible with the closure operator $\overline{(-)}^{\mathbb{T}}$, that is, we have a factorization $\overline{(-)}^{\mathbb{T}}_{\tilde{\mathcal{S}}} : \mathcal{P}(\tilde{\mathcal{S}}) \rightarrow \mathcal{P}(\mathcal{S})$ of $\overline{(-)}^{\mathbb{T}} : \mathcal{P}(\mathcal{S}) \rightarrow \mathcal{P}(\mathcal{S})$ through the image $\mathcal{P}(\mathcal{S}) \rightarrow \mathcal{P}(\tilde{\mathcal{S}})$ under $\mathcal{P}$ of the natural projection map $\mathcal{S} \rightarrow \tilde{\mathcal{S}}$.

The closed points with respect to these closure operators are respectively the syntactic-equivalence classes of quotients of $\mathbb{T}$ and the Grothendieck topologies J on $\mathcal{C}_{\mathbb{T}}$ such that $J \supseteq J_{\mathbb{T}}$.

Let us define $F : \mathcal{P}(\mathcal{S}) \rightarrow \mathcal{P}(S(\mathcal{C}_{\mathbb{T}}))$ as the composite $\overline{(-)}^T \circ \mathcal{P}(\mathcal{F})$ and $G : \mathcal{P}(S(\mathcal{C}_{\mathbb{T}})) \rightarrow \mathcal{P}(\mathcal{S})$ as the composite $\overline{(-)}^{\mathbb{T}}_{\tilde{\mathcal{S}}} \circ \mathcal{P}(\mathcal{G})$.

Given a collection U of geometric sequents over Σ , we define $\mathbb{T}^U$ to be the collection of all the geometric sequents σ over Σ such that $\mathcal{F}(\sigma)$ belongs to $\overline{\mathcal{F}(U)}^T$. Similarly, given a collection V of sieves on $\mathcal{C}_{\mathbb{T}}$, we define J_V to be the collection of sieves R in $\mathcal{C}$ such that any sequent in $\mathcal{G}(R)$ is provable in $\mathbb{T} \cup \mathcal{G}(V)$ (by using geometric logic).

The following result shows that the maps $\mathcal{P}(\mathcal{F})$ and $\mathcal{P}(\mathcal{G})$ are compatible with respect to these closure operators and that F and G are inverse to each other on the subsets of closed points, that is between the collection of syntactic-equivalence classes of quotients of $\mathbb{T}$ and the collection of Grothendieck topologies

on $\mathcal{C}_{\mathbb{T}}$ which contain $J_{\mathbb{T}}$. In fact, given a quotient $\mathbb{T}'$ of $\mathbb{T}$, $F(\mathbb{T}') = J_{\mathbb{T}'}^{\mathbb{T}}$, while, for a Grothendieck topology $J \supseteq J_{\mathbb{T}}$, $G(J) = \mathbb{T}^J$ (using the notations of section 3.2.1). This approach thus provides a different, entirely syntactic, way to arrive at the duality of Theorem 3.2.5.

Theorem 3.2.7 *With the above notation, we have that*

- (i) *For any $U \in \mathcal{P}(\mathcal{S})$, $\mathcal{F}(\overline{U}^{\mathbb{T}}) \subseteq \overline{\mathcal{F}(U)}^{\mathcal{T}}$.*
- (ii) *For any $V \in \mathcal{P}(S(\mathcal{C}_{\mathbb{T}}))$, $\mathcal{G}(\overline{V}^{\mathcal{T}}) \subseteq \overline{\mathcal{G}(V)}^{\mathbb{T}}$.*
- (iii) *For any $U \in \mathcal{P}(\mathcal{S})$, $G(F(U)) = \overline{U}^{\mathbb{T}} = \mathbb{T}^U$.*
- (iv) *For any $V \in \mathcal{P}(S(\mathcal{C}_{\mathbb{T}}))$, $F(G(V)) = \overline{V}^{\mathcal{T}} = J_V$.*

In other words, the maps $\mathcal{P}(F)$ and $\mathcal{P}(G)$ define a proof-theoretic equivalence between the classical deduction system for geometric logic over $\mathbb{T}$ and the proof system $\mathcal{T}_{\mathcal{C}_{\mathbb{T}}}^{J_{\mathbb{T}}}$.

Proof

- (i) We have to prove that, given $U := \{\sigma_i \mid i \in I\} \in \mathcal{P}(\mathcal{S})$, if a geometric sequent σ is provable in $\mathbb{T} \cup U$ using geometric logic, then $\mathcal{F}(\{\sigma\})$ belongs to $\overline{\mathcal{F}(U)}^{\mathcal{T}}$. Let us show this by induction on the complexity of a proof of $\sigma \equiv (\phi \vdash_{\vec{x}} \psi)$ in $\mathbb{T} \cup U$.

If $\sigma \in U$ then the thesis is clear.

If σ belongs to $\mathbb{T}$ or, more generally, is provable in $\mathbb{T}$, then the morphism

$$\{\vec{x}' . \phi \wedge \psi\} \xrightarrow{[\phi \wedge \psi \wedge \vec{x}' = \vec{x}]} \{\vec{x} . \phi\}$$

in $\mathcal{C}_{\mathbb{T}}$ is isomorphic to the identity morphism on $\{\vec{x} . \phi\}$ and hence belongs to $\overline{\mathcal{F}(U)}^{\mathcal{T}}$ by the maximality axiom for Grothendieck topologies. Notice in particular that if σ is an axiom of geometric logic then $\mathcal{F}(\sigma)$ belongs to $\overline{\mathcal{F}(U)}^{\mathcal{T}}$.

Let us now verify that all the inference rules for geometric logic are ‘sound’ with respect to the operation $\mathcal{F}$, i.e. that if each of the premises σ of an inference rule satisfies ‘ $\mathcal{F}(\sigma)$ belongs to $\overline{\mathcal{F}(U)}^{\mathcal{T}}$ ’, then its conclusion σ' also satisfies ‘ $\mathcal{F}(\sigma')$ belongs to $\overline{\mathcal{F}(U)}^{\mathcal{T}}$ ’.

Substitution rule:

$$\frac{(\phi \vdash_{\vec{x}} \psi)}{(\phi[\vec{s}/\vec{x}] \vdash_{\vec{y}} \psi[\vec{s}/\vec{x}])}$$

where $\vec{y}$ is any string of variables including all the variables occurring in a string of terms $\vec{s}$.

We have to prove that if the principal sieve in $\mathcal{C}_{\mathbb{T}}$ generated by the morphism

$$\{\vec{x}' . \phi \wedge \psi\} \xrightarrow{[\phi \wedge \psi \wedge \vec{x}' = \vec{x}]} \{\vec{x} . \phi\}$$

is $\overline{\mathcal{F}(U)}^T$ -covering then the principal sieve generated by the morphism

$$\{\vec{y}' . \phi[\vec{s}/\vec{x}] \wedge \psi[\vec{s}/\vec{x}]\} \xrightarrow{[(\phi[\vec{s}/\vec{x}] \wedge \psi[\vec{s}/\vec{x}] \wedge \vec{y}' = \vec{y})]} \{\vec{y}' . \phi[\vec{s}/\vec{x}]\}$$

is also $\overline{\mathcal{F}(U)}^T$ -covering.

For any geometric formula $\phi(\vec{x})$ and a term $s(\vec{y})$ over Σ , the diagram

$$\begin{array}{ccc} \{\vec{y}' . \phi[\vec{s}/\vec{x}]\} & \xrightarrow{[(s(\vec{y}') = \vec{x}) \wedge \phi]} & \{\vec{x}' . \phi[\vec{x}'/\vec{x}]\} \\ \downarrow [\phi[\vec{s}/\vec{x}] \wedge \vec{y}' = \vec{y}] & & \downarrow [\phi \wedge \vec{x}' = \vec{x}] \\ \{\vec{y}' . \top\} & \xrightarrow{[s(\vec{y}) = \vec{x}]} & \{\vec{x}' . \top\} \end{array}$$

is a pullback in $\mathcal{C}_{\mathbb{T}}$. To prove this, we preliminarily observe that if $\chi \equiv (\exists \vec{y})\xi$ is a geometric formula in a context $\vec{x}$ such that the sequent

$$(\xi \wedge \xi[\vec{z}/\vec{y}]) \vdash_{\vec{x}, \vec{y}, \vec{z}} \vec{y}' = \vec{z}$$

is provable in $\mathbb{T}$ then the objects $\{\vec{x}' . \chi\}$ and $\{\vec{x}' . \vec{y}' . \xi\}$ are isomorphic in $\mathcal{C}_{\mathbb{T}}$. Indeed, it is an easy consequence of Lemma 1.4.2 that the arrow

$$\{\vec{x}' . \vec{y}' . \xi\} \xrightarrow{[\xi \wedge (\vec{x}' = \vec{x})]} \{\vec{x}' . \chi[\vec{x}'/\vec{x}]\}$$

is an isomorphism.

Now, it immediately follows from the substitution rule (and the equality axioms) that the bisequent $((\exists \vec{x})((s(\vec{y}) = \vec{x}) \wedge \phi(\vec{x}))) \dashv\vdash_{\vec{y}} \phi[\vec{s}/\vec{x}]$ is provable in geometric logic.

In view of the construction of pullbacks in $\mathcal{C}_{\mathbb{T}}$ given in section 3.2.1, these two remarks together imply that the above square is a pullback in $\mathcal{C}_{\mathbb{T}}$, as required.

From this fact we immediately deduce that the morphism

$$\{\vec{x}' . \phi[\vec{s}/\vec{x}] \wedge \psi[\vec{s}/\vec{x}]\} \xrightarrow{[\phi[\vec{s}/\vec{x}] \wedge \psi[\vec{s}/\vec{x}] \wedge \vec{x}' = \vec{x}]} \{\vec{x}' . \phi[\vec{s}/\vec{x}]\}$$

is (isomorphic to) the pullback in $\mathcal{C}_{\mathbb{T}}$ along $[(s(\vec{y}) = \vec{x}) \wedge \phi] : \{\vec{y}' . \phi[\vec{s}/\vec{x}]\} \rightarrow \{\vec{x}' . \phi\}$ of the morphism

$$\{\vec{x}' . \phi \wedge \psi\} \xrightarrow{[\phi \wedge \psi \wedge \vec{x}' = \vec{x}]} \{\vec{x}' . \phi\}.$$

We note that, for a Grothendieck topology J on a category $\mathcal{C}$, if the diagram

$$\begin{array}{ccc} a & \longrightarrow & b \\ \downarrow f' & & \downarrow f \\ d & \xrightarrow{h} & c \end{array}$$

is a pullback square in $\mathcal{C}$ then $(f) \in J(c)$ implies $(f') \in J(d)$. Indeed, by the universal property of the pullback, we have $(f') = h^*((f))$. Applying this to our pullback square yields our thesis.

This concludes our proof that the substitution rule is ‘sound’ for the operation $\mathcal{F}$.

Cut rule:

$$\frac{(\phi \vdash_{\vec{x}} \psi)(\psi \vdash_{\vec{x}} \chi)}{(\phi \vdash_{\vec{x}} \chi)}$$

We have to prove that if the principal sieves in $\mathcal{C}_{\mathbb{T}}$ respectively generated by the morphisms

$$\{\vec{x}' . \phi \wedge \psi\} \xrightarrow{[\phi \wedge \psi \wedge \vec{x}' = \vec{x}]} \{\vec{x}' . \phi\}$$

and

$$\{\vec{x}' . \psi \wedge \chi\} \xrightarrow{[\psi \wedge \chi \wedge \vec{x}' = \vec{x}]} \{\vec{x}' . \psi\}$$

are $\overline{\mathcal{F}(U)}^T$ -covering then the principal sieve generated by the morphism

$$\{\vec{x}' . \phi \wedge \chi\} \xrightarrow{[\phi \wedge \chi \wedge \vec{x}' = \vec{x}]} \{\vec{x}' . \phi\}$$

is also $\overline{\mathcal{F}(U)}^T$ -covering.

The diagrams

$$\begin{array}{ccc} \{\vec{x}''' . \phi \wedge \psi \wedge \chi\} & \xrightarrow{[\phi \wedge \psi \wedge \chi \wedge \vec{x}''' = \vec{x}']} & \{\vec{x}'' . \phi \wedge \chi\} \\ \downarrow [\phi \wedge \psi \wedge \chi \wedge \vec{x}''' = \vec{x}'] & & \downarrow [\phi \wedge \chi \wedge \vec{x}'' = \vec{x}] \\ \{\vec{x}' . \phi \wedge \psi\} & \xrightarrow{[\phi \wedge \psi \wedge \vec{x}' = \vec{x}]} & \{\vec{x}' . \phi\} \end{array}$$

and

$$\begin{array}{ccc} \{\vec{x}''' . \phi \wedge \psi \wedge \chi\} & \xrightarrow{[\phi \wedge \psi \wedge \chi \wedge \vec{x}''' = \vec{x}']} & \{\vec{x}'' . \psi \wedge \chi\} \\ \downarrow [\phi \wedge \psi \wedge \chi \wedge \vec{x}''' = \vec{x}'] & & \downarrow [\psi \wedge \chi \wedge \vec{x}'' = \vec{x}] \\ \{\vec{x}' . \phi \wedge \psi\} & \xrightarrow{[\phi \wedge \psi \wedge \vec{x}' = \vec{x}]} & \{\vec{x}' . \psi\} \end{array}$$

are clearly pullback squares in $\mathcal{C}_{\mathbb{T}}$.

By the stability axiom for Grothendieck topologies, the principal sieve generated by the morphism

$$\{x''' \cdot \phi \wedge \psi \wedge \chi\} \xrightarrow{[\phi \wedge \psi \wedge \chi \wedge x''' = x']} \{x' \cdot \phi \wedge \psi\}$$

is $\overline{\mathcal{F}(U)}^T$ -covering, since it is the pullback of the $(\overline{\mathcal{F}(U)}^T)$ -covering) principal sieve generated by the morphism

$$\{x' \cdot \psi \wedge \chi\} \xrightarrow{[\psi \wedge \chi \wedge x' = \vec{x}]} \{\vec{x} \cdot \psi\}$$

along the arrow

$$\{x' \cdot \phi \wedge \psi\} \xrightarrow{[\phi \wedge \psi \wedge x' = \vec{x}]} \{\vec{x} \cdot \psi\}.$$

Since the principal sieve generated by the morphism

$$\{x' \cdot \phi \wedge \psi\} \xrightarrow{[\phi \wedge \psi \wedge x' = \vec{x}]} \{\vec{x} \cdot \phi\}$$

is $\overline{\mathcal{F}(U)}^T$ -covering, we can conclude, by the transitivity axiom for Grothendieck topologies and the fact that the first of the two squares above is a pullback, that the principal sieve generated by the morphism

$$\{x' \cdot \phi \wedge \chi\} \xrightarrow{[\phi \wedge \chi \wedge x' = \vec{x}]} \{\vec{x} \cdot \phi\}$$

is $\overline{\mathcal{F}(U)}^T$ -covering, as required.

Rule for finite conjunction:

$$\frac{(\phi \vdash_{\vec{x}} \psi)(\phi \vdash_{\vec{x}} \chi)}{(\phi \vdash_{\vec{x}} (\psi \wedge \chi))}$$

We have to prove that if the principal sieves in $\mathcal{C}_{\mathbb{T}}$ respectively generated by the morphisms

$$\{x' \cdot \phi \wedge \psi\} \xrightarrow{[\phi \wedge \psi \wedge x' = \vec{x}]} \{\vec{x} \cdot \phi\}$$

and

$$\{x' \cdot \phi \wedge \chi\} \xrightarrow{[\phi \wedge \chi \wedge x' = \vec{x}]} \{\vec{x} \cdot \phi\}$$

are $\overline{\mathcal{F}(U)}^T$ -covering then the principal sieve generated by the morphism

$$\{x' \cdot \phi \wedge (\psi \wedge \chi)\} \xrightarrow{[\phi \wedge (\psi \wedge \chi) \wedge x' = \vec{x}]} \{\vec{x} \cdot \phi\}$$

is also $\overline{\mathcal{F}(U)}^T$ -covering.

We have observed that the diagram

$$\begin{array}{ccc}
 \{x''' \cdot \phi \wedge \psi \wedge \chi\} & \xrightarrow{[\phi \wedge \psi \wedge \chi \wedge x''' = x']} & \{x'' \cdot \phi \wedge \chi\} \\
 \downarrow [\phi \wedge \psi \wedge \chi \wedge x''' = x'] & & \downarrow [\phi \wedge \chi \wedge x'' = \vec{x}] \\
 \{x' \cdot \phi \wedge \psi\} & \xrightarrow{[\phi \wedge \psi \wedge x' = \vec{x}]} & \{\vec{x} \cdot \phi\}
 \end{array}$$

is a pullback square in $\mathcal{C}_{\mathbb{T}}$. The stability axiom for Grothendieck topologies then ensures that the principal sieve generated by the morphism

$$\{x''' \cdot \phi \wedge \psi \wedge \chi\} \xrightarrow{[\phi \wedge \psi \wedge \chi \wedge x''' = x']} \{x' \cdot \phi \wedge \psi\}$$

is $\overline{\mathcal{F}(U)}^T$ -covering, as it is the pullback of the $(\overline{\mathcal{F}(U)}^T)$ -covering principal sieve generated by the morphism

$$\{x'' \cdot \phi \wedge \chi\} \xrightarrow{[\phi \wedge \chi \wedge x'' = \vec{x}]} \{\vec{x} \cdot \phi\}$$

along the arrow

$$\{x' \cdot \phi \wedge \psi\} \xrightarrow{[\phi \wedge \psi \wedge x' = \vec{x}]} \{\vec{x} \cdot \phi\}.$$

But the principal sieve generated by the morphism

$$\{x' \cdot \phi \wedge \psi\} \xrightarrow{[\phi \wedge \psi \wedge x' = \vec{x}]} \{\vec{x} \cdot \phi\}$$

is $\overline{\mathcal{F}(U)}^T$ -covering and hence, since the morphism

$$\{x' \cdot \phi \wedge (\psi \wedge \chi)\} \xrightarrow{[\phi \wedge (\psi \wedge \chi) \wedge x' = \vec{x}]} \{\vec{x} \cdot \phi\}$$

is equal to the composite of

$$\{x' \cdot \phi \wedge \psi\} \xrightarrow{[\phi \wedge \psi \wedge x' = \vec{x}]} \{\vec{x} \cdot \phi\}$$

and

$$\{x''' \cdot \phi \wedge \psi \wedge \chi\} \xrightarrow{[\phi \wedge \psi \wedge \chi \wedge x''' = x']} \{x' \cdot \phi \wedge \psi\},$$

it follows from property (d) in Definition 3.1.3 that the principal sieve generated by the morphism

$$\{x' \cdot \phi \wedge (\psi \wedge \chi)\} \xrightarrow{[\phi \wedge (\psi \wedge \chi) \wedge x' = \vec{x}]} \{\vec{x} \cdot \phi\}$$

is $\overline{\mathcal{F}(U)}^T$ -covering, as required.

Rule for infinitary disjunction:

$$\frac{\{(\phi_i \vdash_{\vec{x}} \chi) \mid i \in I\}}{\left(\bigvee_{i \in I} \phi_i \vdash_{\vec{x}} \chi\right)}$$

We have to prove that if each of the principal sieves in $\mathcal{C}_{\mathbb{T}}$ generated by the morphisms

$$\{\vec{x}' . \phi_i \wedge \chi\} \xrightarrow{[\phi_i \wedge \chi \wedge \vec{x}' = \vec{x}]} \{\vec{x}' . \phi_i\}$$

(as i varies in I) is $\overline{\mathcal{F}(U)}^T$ -covering then the principal sieve generated by the morphism

$$\{\vec{x}' . \left(\bigvee_{i \in I} \phi_i\right) \wedge \chi\} \xrightarrow{\left[\left(\bigvee_{i \in I} \phi_i\right) \wedge \chi \wedge \vec{x}' = \vec{x}\right]} \{\vec{x}' . \bigvee_{i \in I} \phi_i\}$$

is also $\overline{\mathcal{F}(U)}^T$ -covering.

The sieve on $\{\vec{x}' . \bigvee_{i \in I} \phi_i\}$ generated by the morphisms

$$j_i := \{\vec{x}' . \phi_i\} \xrightarrow{[\phi_i \wedge \vec{x}' = \vec{x}]} \{\vec{x}' . \bigvee_{i \in I} \phi_i\}$$

(as i varies in I) is $\overline{\mathcal{F}(U)}^T$ -covering by definition of $J_{\mathbb{T}}$, since $\overline{\mathcal{F}(U)}^T \supseteq J_{\mathbb{T}}$.

Now, for each $i \in I$, the diagram

$$\begin{array}{ccc} \{\vec{x}''' . \phi_i \wedge \chi\} & \xrightarrow{[\phi_i \wedge \chi \wedge \vec{x}''' = \vec{x}']} & \{\vec{x}'' . \left(\bigvee_{i \in I} \phi_i\right) \wedge \chi\} \\ \downarrow [\phi_i \wedge \chi \wedge \vec{x}''' = \vec{x}'] & & \downarrow \left[\left(\bigvee_{i \in I} \phi_i\right) \wedge \chi \wedge \vec{x}'' = \vec{x}\right] \\ \{\vec{x}' . \phi_i\} & \xrightarrow{j_i} & \{\vec{x}' . \bigvee_{i \in I} \phi_i\} \end{array}$$

is a pullback square in $\mathcal{C}_{\mathbb{T}}$. Our thesis thus follows from the transitivity axiom for Grothendieck topologies.

Rules for existential quantification:

$$\frac{(\phi \vdash_{\vec{x}, \vec{y}} \psi)}{((\exists \vec{y}) \phi \vdash_{\vec{x}} \psi)}$$

where $\vec{y}$ is not free in ψ .

We have to prove that the principal sieve in $\mathcal{C}_{\mathbb{T}}$ generated by the morphism

$$\{\vec{x}', \vec{y}' . \phi \wedge \psi\} \xrightarrow{[\phi \wedge \psi \wedge \vec{x}' = \vec{x} \wedge \vec{y}' = \vec{y}]} \{\vec{x}, \vec{y} . \phi\}$$

is $\overline{\mathcal{F}(U)}^T$ -covering if and only if the principal sieve generated by the morphism

$$\{\vec{x}' . ((\exists \vec{y})\phi) \wedge \psi\} \xrightarrow{[(\exists \vec{y})\phi] \wedge \vec{x}' = \vec{x}} \{\vec{x}' . (\exists \vec{y})\phi\}$$

is $\overline{\mathcal{F}(U)}^T$ -covering.

The diagram

$$\begin{array}{ccc} \{x'', y'' . (\phi \wedge \psi)[x''/\vec{x}, y''/\vec{y}]\} & \xrightarrow{[\phi \wedge \psi \wedge x'' = \vec{x}' \wedge y'' = \vec{y}']} & \{x'', y'' . \phi[x''/\vec{x}, y''/\vec{y}]\} \\ \downarrow [\phi \wedge \psi \wedge x'' = \vec{x}'] & & \downarrow [\phi \wedge x'' = \vec{x}] \\ \{x' . ((\exists \vec{y})\phi \wedge \psi)[x'/\vec{x}]\} & \xrightarrow{[(\exists \vec{y})\phi] \wedge \psi \wedge x' = \vec{x}} & \{\vec{x} . (\exists \vec{y})\phi\} \end{array}$$

is a pullback square in $\mathcal{C}_{\mathbb{T}}$ (this follows at once from the construction of pullbacks in $\mathcal{C}_{\mathbb{T}}$ reviewed in section 3.2.1 by invoking the rules for existential quantification, as in the proof for the substitution rule).

The ‘if’ part of our claim clearly follows from the stability axiom for Grothendieck topologies, so it remains to prove the ‘only if’ part. To this end, we notice that the morphism

$$\{x'', y'' . \phi[x''/\vec{x}, y''/\vec{y}]\} \xrightarrow{[\phi \wedge x'' = \vec{x}]} \{\vec{x} . (\exists \vec{y})\phi\}$$

is a cover in $\mathcal{C}_{\mathbb{T}}$; this implies that the sieve generated by it is $\overline{\mathcal{F}(U)}^T$ -covering by definition of $J_{\mathbb{T}}$, since $\overline{\mathcal{F}(U)}^T \supseteq J_{\mathbb{T}}$. Hence, since the above square is cartesian, the principal sieve generated by the morphism

$$\{\vec{x}' . ((\exists \vec{y})\phi) \wedge \psi\} \xrightarrow{[(\exists \vec{y})\phi] \wedge \psi \wedge \vec{x}' = \vec{x}} \{\vec{x}' . (\exists \vec{y})\phi\}$$

is $\overline{\mathcal{F}(U)}^T$ -covering by properties (b) and (d) in Definition 3.1.3.

This completes the proof of part (i) of the theorem.

- (ii) We have to prove that, given $V \in \mathcal{P}(S(\mathcal{C}_{\mathbb{T}}))$, if a sieve R is $\overline{V}^T$ -covering then any sequent in $\mathcal{G}(R)$ is provable in $\mathbb{T} \cup \mathcal{G}(V)$ using geometric logic,

that is $J_V \supseteq \bar{V}^T$. In fact, we will show that J_V is a Grothendieck topology containing $J_{\mathbb{T}}$ and all the sieves in V as its covering sieves; this will immediately imply our thesis.

By definition of J_V , the sieves in V are J_V -covering, and if R is a $J_{\mathbb{T}}$ -covering sieve then, by definition of $J_{\mathbb{T}}$, any sequent in $\mathcal{G}(R)$ is provable in $\mathbb{T}$, whence R is J_V -covering. To prove that J_V is a Grothendieck topology, we use Definition 3.1.3. Property (a) is obvious and property (b) easily follows from the cut rule in geometric logic. Property (c) is an immediate consequence of the observation following the first cartesian square in the proof of Theorem 3.2.5. It remains to prove property (d). Since $\mathcal{G}(R) = \mathcal{G}(\bar{R}^{J_{\mathbb{T}}})$ for any sieve R in $\mathcal{C}_{\mathbb{T}}$ (by Proposition 3.1.6 and Remark 3.1.7), it suffices to prove that for any sieve S in $\mathcal{C}_{\mathbb{T}}$ generated by a monomorphism $m : d \rightarrow c$ and any sieve T in $\mathcal{C}_{\mathbb{T}}$ on d , if both S and T are J_V -covering then $S * \{T\}$ is J_V -covering. Now, in view of the equality $\mathcal{G}(R) = \mathcal{G}(\bar{R}^{J_{\mathbb{T}}})$, our thesis easily follows from the cut rule in geometric logic, by using Proposition 3.1.6, Remark 3.1.7 and Remark 3.1.4.

- (iii) Let us first show that $\overline{\mathcal{G}(\mathcal{F}(U))}^{\mathbb{T}} = \bar{U}^{\mathbb{T}}$. Note that $\mathcal{G}(\mathcal{F}(U))$ is the collection of sequents of the form $\mathcal{G}(\mathcal{F}(\sigma))$ as σ varies in U . If $\sigma \equiv (\phi \vdash_{\bar{x}} \psi)$ then $\mathcal{G}(\mathcal{F}(\sigma))$ is the $\mathbb{T}$ -equivalence class of the sequent $(\phi \vdash_{\bar{x}} \phi \wedge \psi)$; but this sequent is clearly $\mathbb{T}$ -equivalent to σ , whence $\overline{\mathcal{G}(\mathcal{F}(U))}^{\mathbb{T}} = \bar{U}^{\mathbb{T}}$, as required.

We have that

$$G(F(U)) = \overline{\mathcal{G}(F(U))}^{\mathbb{T}} = \overline{\overline{\mathcal{G}(\mathcal{F}(U))}^{\mathbb{T}}}^{\mathbb{T}} = \overline{\overline{\mathcal{G}(\mathcal{F}(U))}^{\mathbb{T}}}^{\mathbb{T}} = \overline{\mathcal{G}(\mathcal{F}(U))}^{\mathbb{T}} = \bar{U}^{\mathbb{T}},$$

where the third equality follows from part (i) of the theorem. To complete the proof of part (iii) of the theorem, it thus remains to show that $\bar{U}^{\mathbb{T}} = \mathbb{T}^U$. The inclusion $\bar{U}^{\mathbb{T}} \subseteq \mathbb{T}^U$ follows from part (i) of the theorem, while the converse inclusion follows as a consequence of the equality $\overline{\mathcal{G}(\mathcal{F}(U))}^{\mathbb{T}} = \bar{U}^{\mathbb{T}}$ and of part (i) of the theorem: if $\sigma \in \mathbb{T}^U$ then $\sigma \in \{\sigma\}^{\mathbb{T}} = \overline{\mathcal{G}(\mathcal{F}(\{\sigma\}))}^{\mathbb{T}} \subseteq \overline{\mathcal{G}(\mathcal{F}(U))}^{\mathbb{T}} = \bar{U}^{\mathbb{T}}$.

- (iv) Let us first prove that $\overline{\mathcal{F}(\mathcal{G}(V))}^T = \bar{V}^T$. We observe that $\mathcal{F}(\mathcal{G}(V))$ is the collection of sieves of the form $\mathcal{F}(\mathcal{G}(R))$ as R varies in V . Moreover, we have that $\mathcal{F}(\mathcal{G}(R)) = \bar{R}^{J_{\mathbb{T}}}$. Our claim thus follows from Proposition 3.1.5. Now, by using the fact that $\overline{\mathcal{F}(\mathcal{G}(V))}^T = \bar{V}^T$, one can prove the required equalities as in the proof of part (iii) of the theorem, with part (ii) playing here the role of part (i) there.

□

Remark 3.2.8 Let $\mathbb{T}$ be a geometric theory over a signature Σ . Given a quotient $\mathbb{T}'$ of $\mathbb{T}$, let $J_{\mathbb{T}'}^{\mathbb{T}}$ be the associated $\mathbb{T}$ -topology of $\mathbb{T}'$. Then the equalities $\bar{U}^{\mathbb{T}} = \mathbb{T}^U$ and $\bar{V}^{\mathbb{T}} = J_V$ of Theorem 3.2.7 yield the following equivalences:

- (i) For any sieve $R \in S(\mathcal{C}_{\mathbb{T}})$, $R \in J_{\mathbb{T}'}^{\mathbb{T}}$ if and only if any sequent in $\mathcal{G}(R)$ is provable in $\mathbb{T}'$.
- (ii) For any geometric sequent σ over Σ , σ is provable in $\mathbb{T}'$ if and only if $\mathcal{F}(\sigma)$ is $J_{\mathbb{T}'}^{\mathbb{T}}$ -covering.

In particular, we obtain the following characterization of the syntactic topology $J_{\mathbb{T}}$ on $\mathcal{C}_{\mathbb{T}}$: a sieve R is $J_{\mathbb{T}}$ -covering if and only if some (equivalently, every) sequent in $\mathcal{G}(R)$ is provable in $\mathbb{T}$.

The interest of Theorem 3.2.7 lies in the fact that the proof system $\mathcal{T}_{\mathcal{C}_{\mathbb{T}}}^{J_{\mathbb{T}}}$ is, for many purposes, computationally better behaved than the classical deduction system for geometric logic over $\mathbb{T}$; indeed, it is often easier to determine whether a given sieve belongs to the Grothendieck topology generated by a certain family of sieves than to test whether a given geometric sequent is provable in a certain geometric theory.

Let us briefly consider how much of Theorem 3.2.7 survives for smaller fragments of geometric logic, e.g. cartesian, regular, or coherent logic. If $\mathbb{T}$ is a cartesian (resp. regular, coherent) theory over Σ , one can define exactly as above an assignment $\mathcal{F}$ from the collection $\mathcal{S}^{\text{cart}}$ (resp. $\mathcal{S}^{\text{reg}}$, $\mathcal{S}^{\text{coh}}$) of cartesian (resp. regular, coherent) sequents over Σ to the collection $S(\mathcal{C}_{\mathbb{T}}^{\text{cart}})$ (resp. $S(\mathcal{C}_{\mathbb{T}}^{\text{reg}})$, $S(\mathcal{C}_{\mathbb{T}}^{\text{coh}})$) of sieves in the cartesian (resp. regular, coherent) syntactic category of the theory $\mathbb{T}$. Accordingly, the closure operator on $\mathcal{P}(\mathcal{S}^{\text{cart}})$ (resp. $\mathcal{P}(\mathcal{S}^{\text{reg}})$, $\mathcal{P}(\mathcal{S}^{\text{coh}})$) sends a collection U of sequents in $\mathcal{S}^{\text{cart}}$ (resp. $\mathcal{S}^{\text{reg}}$, $\mathcal{S}^{\text{coh}}$) to the collection of cartesian (resp. regular, coherent) sequents over Σ which are derivable in $U \cup \mathbb{T}$ by using cartesian (resp. regular, coherent) logic, and it is immediate to see that the proof of part (i) of the theorem continues to hold. On the contrary, no assignment $\mathcal{G}$ with values in the class of cartesian (resp. regular, coherent) sequents over Σ can be defined, since one should restrict to sieves generated by a monomorphism (resp. a single arrow, a finite number of arrows); however, if we consider $\mathcal{G}$ to take values in the class of geometric sequents over Σ as in the geometric case then part (ii) of the theorem continues to hold and for any pre-sieve V in the cartesian (resp. regular, coherent) category $\mathcal{C}_{\mathbb{T}}^{\text{cart}}$ (resp. $\mathcal{C}_{\mathbb{T}}^{\text{reg}}$, $\mathcal{C}_{\mathbb{T}}^{\text{coh}}$), the theory $\mathcal{G}(\bar{V}^{\mathbb{T}})$ is classified by the topos $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\text{cart}}, \bar{V}^{\mathbb{T}})$ (resp. $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\text{reg}}, \bar{V}^{\mathbb{T}})$, $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\text{coh}}, \bar{V}^{\mathbb{T}})$) – cf. the proof of Theorem 3.2.5.

Remarks 3.2.9 (a) Given a geometric quotient $\mathbb{T}'$ of $\mathbb{T}$, it is natural to look for axiomatizations of $\mathbb{T}'$ over $\mathbb{T}$ which are ‘as simple as possible’; this translates, via Theorem 3.2.5, into the problem of finding an ‘as simple as possible’ set of generators for the associated Grothendieck topology $J_{\mathbb{T}'}^{\mathbb{T}}$ over $J_{\mathbb{T}}$; in fact, if a collection V of presieves in $\mathcal{C}_{\mathbb{T}}$ generates a Grothendieck topology J , then, by Theorem 3.2.7(ii), $\mathbb{T}^J$ is axiomatized over $\mathbb{T}$ by the collection of sequents

in $\mathcal{G}(V)$. Note that, conversely, if a collection U of geometric sequents axiomatizes a quotient $\mathbb{T}'$ then, by Theorem 3.2.7(i), the collection of presieves $\mathcal{F}(U)$ generates over $J_{\mathbb{T}}$ the Grothendieck topology $J_{\mathbb{T}'}^{\mathbb{T}}$. For instance, one may ask if $\mathbb{T}'$ can be axiomatized over $\mathbb{T}$ by geometric sequents of the form $(\top \vdash_{\vec{x}} \phi)$; this corresponds to requiring that $J_{\mathbb{T}'}^{\mathbb{T}}$ should be generated over $J_{\mathbb{T}}$ by a collection of principal sieves generated by subobjects of objects of the form $\{\vec{x}. \top\}$; two notable classes of theories with this property are given by the Booleanizations and DeMorganizations of geometric theories (cf. Remark 3.2.9(b) and section 4.2.3 below).

- (b) One can often get interesting insights on theories by investigating the associated Grothendieck topologies. As an illustration of this, let us show that the Booleanization $\mathbb{T}'$ of a geometric theory $\mathbb{T}$, i.e. the quotient of $\mathbb{T}$ corresponding via the duality of Theorem 3.2.5 to the subtopos $\mathbf{sh}_{\neg\neg}(\mathbf{Set}[\mathbb{T}]) \hookrightarrow \mathbf{Set}[\mathbb{T}]$ (see section 1.3.4 above), is axiomatized over $\mathbb{T}$ by geometric sequents of the form $(\top \vdash_{\vec{x}} \phi)$ as claimed in (a). Given a Heyting category $\mathcal{C}$, let $\tilde{\mathcal{C}}$ be its full subcategory on the non-zero objects. Since the only objects which are $J_{\mathbb{T}}$ -covered by the empty sieve are the zero ones, $\tilde{\mathcal{C}}_{\mathbb{T}}$ is $J_{\mathbb{T}}$ -dense in $\mathcal{C}_{\mathbb{T}}$ and the induced Grothendieck topology $J_{\mathbb{T}}|_{\tilde{\mathcal{C}}_{\mathbb{T}}}$ is contained in the double negation topology on $\tilde{\mathcal{C}}_{\mathbb{T}}$; hence $J_{\mathbb{T}'}^{\mathbb{T}}$ is generated over $J_{\mathbb{T}}$ by the sieves generated in $\mathcal{C}_{\mathbb{T}}$ by the principal stably non-empty sieves in $\tilde{\mathcal{C}}_{\mathbb{T}}$ (cf. Proposition 4.2.11 below). Now, given a Heyting category $\mathcal{C}$ and monomorphisms $f : d \rightarrow c'$ and $g : c' \rightarrow c$ in $\tilde{\mathcal{C}}$, it is immediate to see that if (f) , regarded as a sieve in $\tilde{\mathcal{C}}$, is stably non-empty then $f \cong g^*((g \circ f) \cup \neg(g \circ f))$ where $\cup$ and $\neg$ are respectively the union and pseudocomplementation in the Heyting algebra $\text{Sub}_{\mathcal{C}}(c)$, whence (f) is the pullback of a stably non-empty sieve in $\tilde{\mathcal{C}}$ on c . Therefore, since every object in $\mathcal{C}_{\mathbb{T}}$ admits a monomorphism to an object of the form $\{\vec{x}. \top\}$, the topology $J_{\mathbb{T}'}^{\mathbb{T}}$ is generated over $J_{\mathbb{T}}$ by a collection of principal sieves generated by subobjects of objects of the form $\{\vec{x}. \top\}$, as required.

3.3 A deduction theorem for geometric logic

The following result represents an analogue for geometric logic of the deduction theorem in classical first-order logic; we shall derive it as an application of our proof-theoretic interpretation of Theorem 3.2.5.

Theorem 3.3.1 *Let $\mathbb{T}$ be a geometric theory over a signature Σ and ϕ, ψ geometric sentences over Σ such that the sequent $(\top \vdash_{\square} \psi)$ is provable in the theory $\mathbb{T} \cup \{(\top \vdash_{\square} \phi)\}$. Then the sequent $(\phi \vdash_{\square} \psi)$ is provable in the theory $\mathbb{T}$.*

Proof By Theorem 3.2.5 and Lemma 1.4.2 [50], we can rephrase our thesis as follows: if the principal sieve in $\mathcal{C}_{\mathbb{T}}$ generated by the arrow $\{\square. \psi\} \xrightarrow{[\psi]} \{\square. \top\}$ belongs to the Grothendieck topology on $\mathcal{C}_{\mathbb{T}}$ generated over $J_{\mathbb{T}}$ by the principal sieve generated by the arrow $\{\square. \phi\} \xrightarrow{[\phi]} \{\square. \top\}$, then $[\phi] \leq [\psi]$ in $\text{Sub}_{\mathcal{C}_{\mathbb{T}}}(\{\square. \top\})$.

Recalling Theorem 1.4.13 and the fact that the syntactic topology $J_{\mathbb{T}}$ is the geometric topology on the category $\mathcal{C}_{\mathbb{T}}$, we can further rewrite our thesis as follows: if $\mathcal{C}$ is a geometric category and $J_{\mathcal{C}}^{\text{geom}}$ is the geometric topology on it then, given subobjects $m : a \rightarrow 1$ and $n : b \rightarrow 1$ of the terminal object 1 of $\mathcal{C}$ such that (n) belongs to the Grothendieck topology generated over $J_{\mathcal{C}}^{\text{geom}}$ by the sieve (m) , $m \leq n$ in $\text{Sub}_{\mathcal{C}}(1)$.

Let us apply the formula for the Grothendieck topology $(D^r)^l$ generated by a family of sieves D that is stable under pullback (cf. section 4.1.1 below) to the collection D of sieves which are either $J_{\mathcal{C}}^{\text{geom}}$ -covering or of the form $f^*((m))$ for an arrow f with codomain 1; starting from the assumption that $(n) \in (D^r)^l(b)$, we want to deduce that $m \leq n$ in $\text{Sub}_{\mathcal{C}}(1)$.

We note that $m \leq n$ if and only if $m \leq (m \Rightarrow n)$, if and only if $m^*(m \Rightarrow n) \cong 1_a$ (where $\Rightarrow$ denotes the Heyting implication in $\text{Sub}_{\mathcal{C}}(1)$ – note that this operation is well-defined by Proposition 1.3.14).

Now, by the results of section 4.1.1, we have that for any $c \in \mathcal{C}$

$$(D^r)^l(c) = \{S \text{ sieve on } c \mid \text{for any arrow } d \xrightarrow{f} c \text{ and sieve } Z \text{ on } d, \\ [Z \in D^r(d) \text{ and } f^*(S) \subseteq Z] \text{ implies } Z = M_d\}.$$

Taking Z to be $m^*((m \Rightarrow n))$, S to be (n) and f to be m in this formula, we see that, since $n \leq (m \Rightarrow n)$, in order to prove that $m^*(m \Rightarrow n) \cong 1_a$ it suffices to show that $m^*((m \Rightarrow n)) \in D^r(a)$. In fact, we shall prove that $(m \Rightarrow n) \in D^r(1)$, which implies that $m^*((m \Rightarrow n)) \in D^r(a)$ since D^r is stable under pullback.

Again by the results of section 4.1.1, we have that for any $c \in \mathcal{C}$

$$D^r(c) = \{T \text{ sieve on } c \mid \text{for any arrow } d \xrightarrow{f} c \text{ and sieve } S \text{ on } d, \\ [S \in D(d) \text{ and } S \subseteq f^*(T)] \text{ implies } f \in T\}.$$

To show that $(m \Rightarrow n) \in D^r(1)$, we are thus reduced to prove that for any arrow $f : d \rightarrow 1$ with codomain 1 and any sieve S on d such that $S \in D(d)$, $S \subseteq f^*((m \Rightarrow n))$ implies $1_d \in f^*((m \Rightarrow n))$. Now, if $S \in D(d)$ then there are two possible cases: either S is $J_{\mathcal{C}}^{\text{geom}}$ -covering or (since 1 is a terminal object) S is equal to $f^*((m))$. In the first case, we have that if $S \subseteq f^*((m \Rightarrow n))$ then $f^*((m \Rightarrow n))$ is $J_{\mathcal{C}}^{\text{geom}}$ -covering and hence, it being generated by a monomorphism, maximal, as required. In the second case, we have that if $S \subseteq f^*((m \Rightarrow n))$ then $f^*(m) \leq f^*(m \Rightarrow n)$; but $f^*(m \Rightarrow n) = f^*(m) \Rightarrow f^*(n)$ (cf. Lemma A1.4.13 [55]) whence $f^*(m) \leq f^*(m \Rightarrow n)$ implies $f^*(m) \leq f^*(n)$ i.e. $1_d \in f^*((m \Rightarrow n))$. $\square$